5

Presenting the Past in the Digital Age

在数字时代呈现过去

At the heart of digital history, since the early 1990s, there has been an endeavour to utilize the opportunities created by information technology, not only for the study of the past but also for presenting it both in the classroom and before the general public. When Daniel J. Cohen and Roy Rosenzweig published their *Digital History: A Guide to Gathering, Preserving, and Presenting the Past on the Web* in 2006, its introduction stated that the book had emerged in response to the huge expansion of historical content on the World Wide Web: 'Virtually every historical archive, historical society, historic house, and historic site - even the very smallest - has its own website. So does just about every re-enactment group, genealogical society, and body of historical enthusiasts.'[1] The change from the early 1990s to 2006 was really significant: digital resources had multiplied. Simultaneously, the use of information technology by historians had become a matter of course, or at least a dimension that could not be neglected.

自20世纪90年代初以来，数字史的核心一直包含这样一种努力：利用信息技术创造的机会，不仅研究过去，也在课堂和公众面前呈现过去。2006年，Daniel J. Cohen 与 Roy Rosenzweig 出版《Digital History: A Guide to Gathering, Preserving, and Presenting the Past on the Web》时，在导言中指出，这本书正是为了回应万维网上历史内容的大幅增长而写成的："几乎每一个历史档案馆、历史学会、历史建筑和历史遗址——哪怕规模再小——都有自己的网站。几乎每一个历史重演团体、族谱学会和历史爱好者组织也是如此。"[1] 从20世纪90年代初到2006年的变化确实十分显著：数字资源成倍增长。与此同时，历史学家使用信息技术也已成为一种常态，至少已经成为一个无法忽视的研究维度。

This chapter returns to digital historians' interest in using digital tools to present the past. It continues by discussing digital history as public history, then pays attention to the possibilities of visualizing data. The chapter ends with technologies known as augmented and mixed realities and their potential for the digital historian.

本章重新讨论数字史学者利用数字工具呈现过去的兴趣。接下来首先把数字史作为公共史加以讨论，然后关注数据可视化的各种可能性。最后，本章介绍增强现实与混合现实等技术，并讨论它们对数字史研究者可能具有的意义。

Digital history for the public

面向公众的数字史

Since Cohen and Rosenzweig's 2006 book, the resources have further expanded to such an extent that it would be impossible to write a single guidebook on the historical sources and services that are available. If, in the early years of the World Wide Web, the user of the Internet had to be acquainted with HTML code to start his or her home page for a history project, today there are easy-to-use templates offered by WordPress, Blogspot

and other services and, therefore, an abundant amount of content.

自Cohen和Rosenzweig在2006年出版那本书以来，数字资源又进一步扩张，以至于如今几乎不可能再用一本指南完整介绍现有的历史资料与服务。万维网早期，如果网络使用者想为自己的历史项目建立主页，还必须熟悉HTML代码；而今天，WordPress、Blogspot等服务提供了易于使用的模板，因此网络上的相关内容已经极为丰富。

Recent decades have seen the birth of databases that offer metadata and historical material, from archaeological excavations to ancient and modern classics and numerous other themes. For example, in 2010 researchers from the universities of Cambridge and Umeå published their findings on the Jewish communities in the Byzantine Empire as an open database for other scholars and the public.[2] Another interesting project has been *The Texas Slavery Project* by Andrew J. Torget. It started in 2007 and examined 'the spread of American slavery into the borderlands between the United States and Mexico in the decades between 1820 and 1850'. Besides dynamic, interactive maps that describe the flows of enslaved and slaveholder populations in Texas, it offers a population database that allows the user to make discoveries in the history of slavery.[3] Torget's database grew out of a dissertation project and deepened into a digital platform that can be consulted by teachers, students and other researchers.[4]

近几十年来，大量数据库相继出现，它们提供元数据和历史材料，主题从考古发掘、古代与近代经典一直延伸到众多其他领域。例如，2010年，剑桥大学和于默奥大学的研究者把他们关于拜占庭帝国犹太社群的研究成果制作成开放数据库，供其他学者和公众使用。[2]
另一个有意思的项目是Andrew J. Torget主持的《The Texas Slavery Project》。该项目始于2007年，研究“1820年至1850年间美国奴隶制如何扩展到美国与墨西哥之间的边境地区”。除了用动态交互地图展示得克萨斯州被奴役人口和奴隶主人口的流动外，项目还提供人口数据库，使使用者能够据此探索奴隶制历史。[3] Torget的数据库最初源于一个博士论文项目，后来逐渐发展成教师、学生和其他研究者都可以使用的数字平台。[4]

It combines digital and public history successfully. *The Texas Slavery Project* started as a bottom-up project, from the passion to contribute to the growing field of digital resources. There are numerous examples of this, but let us mention only one project that started among a group of undergraduate students at the National University of Colombia in 2015 and 2016. As a product of university courses on digital history and medieval history, they established a website entitled Lorenzetti Digital, which has published both images and analyses of the frescos of the Florentine artist Ambrogio Lorenzetti (1290-1348). Here, digital history appears as an endeavour for presenting historical knowledge via the Internet in a form that blends textual information with visuality.[5]

它成功地把数字史与公共史结合起来。《The Texas Slavery Project》最初是一个自下而上的项目，动力来自参与不断扩大的数字资源领域的热情。类似的例子还有很多，这里只举一个：2015年至2016年间，哥伦比亚国立大学的一群本科生发起了一个项目。这个项目源于大学里的数字史与中世纪史课程，他们建立了名为Lorenzetti Digital的网站，发布佛罗伦萨艺术家Ambrogio Lorenzetti（1290–1348）壁画的图像与分析。在这里，数字史体现为一种通过互联网呈现历史知识的实践，并把文字信息与视觉形式结合起来。[5]

Obviously, digital history does not live only on the Internet, since historians draw on information technology in their work as teachers and public lecturers. Especially in the 2000s, the data projector became like the closest friend for the travelling historian. During recent decades, PowerPoint has been the most popular tool for individual presentations at conferences and in classrooms. The first version of PowerPoint was introduced in 1987 to produce overhead transparencies, but the era of transparencies was soon over. In the early 1990s, universities still had projectors for 35-mm slides, but gradually they were replaced by digital projectors, which

allowed the use of the computer for presentations. Finally, the third version of PowerPoint in 1992 introduced an output for video signal, thus allowing digital slideshows.[6] For digital historians, this introduced the possibility of incorporating words and maps, still and moving images into a lecture presentation, and classroom history got a multimodal character. Eugene Ch'ng and Vincent L. Gaffney have pointed out that 'visualization technologies provide powerful tools that can invoke the sense of presence'.[7] True, digital history can be used as a way to give an audience not only a sense of presence but also a sensory experience that transforms how history is conceived.

当然，数字史并不只存在于互联网中，因为历史学家在教学和面向公众的讲座中同样会使用信息技术。尤其到了21世纪头十年，数据投影仪几乎成了四处讲学的历史学家最亲密的伙伴。近几十年来，PowerPoint一直是学术会议和课堂个人演示中最常用的工具。PowerPoint的第一个版本于1987年推出，最初用于制作投影用透明胶片，但透明胶片的时代很快结束。20世纪90年代初，大学里仍常见35毫米幻灯片投影仪，但它们逐渐被数字投影仪取代，使计算机能够直接用于演示。1992年，PowerPoint第三版加入视频信号输出功能，从而使数字幻灯演示成为可能。[6] 对数字史学者而言，这意味着文字、地图、静态图像和动态图像都可以被整合进讲座中，课堂中的历史呈现也因此具有了多模态特征。Eugene Ch' ng与Vincent L. Gaffney指出，"可视化技术提供了强有力的工具，能够唤起在场感"。[7] 的确，数字史不仅可以让观众产生某种身临其境的感觉，也可以创造一种感官体验，进而改变人们理解历史的方式。

Visualizing data

数据可视化

In our visual world, it is increasingly important for historians to be able to speak the same language as our times, that is, to be able to interpret research results, whatever the theme might be, in visual terms. Visualization is a strategy for organizing and analysing data, both visual or non-visual, but it is also a strategy for presenting the results. To draw on the history of the digital humanities, this was already on the agenda of Franco Moretti in his idea of distant reading and his approach to literary history, which in turn has had a profound impact on the digital scholarship in the 2000s. Therefore, this chapter proceeds by discussing first the question of visualization in a scholarly context and then continues by considering the ways in which visual and audio-visual technologies have been employed. These are both aspects of how the past is presented through digital technologies.

在当下这个高度视觉化的世界中，历史学家越来越需要能够使用与时代相同的"语言"，也就是说，无论研究主题是什么，都要能够用视觉形式解释研究结果。可视化既是一种组织与分析数据的策略——无论数据本身是否具有视觉性——也是一种呈现研究结果的策略。从数字人文的发展史来看，Franco Moretti提出"远读"以及他研究文学史的方式时，就已经把这一问题纳入议程，而这种方法又对21世纪以来的数字学术产生了深远影响。因此，本章先在学术语境中讨论可视化问题，然后继续考察视觉与视听技术的实际运用方式。这两方面都关系到人们如何借助数字技术呈现过去。

For Moretti, visualizations served as a means to synthesize those observations that had been reached through computational methods, as he maintained in his book *Graphs, Maps, Trees: Abstract Models for a Literary History* in 2005. Moretti's argument was that 'deliberate reduction and abstraction' is necessary and, inspired by other fields of research, he had taken 'graphs from quantitative history, maps from geography and trees from evolutionary theory'.[8] Of course, graphs have always been part of the toolbox of economic and social historians who used quantitative methods. This aspect, and the need to draw on statistical expertise, was particularly popular among historians in the 1960s and 1970s. In that sense, many quantitative historians might feel that

there is an affinity with this older tradition and new digital history. For Moretti, this was something different however, especially the way he saw statistics in the service of abstraction.

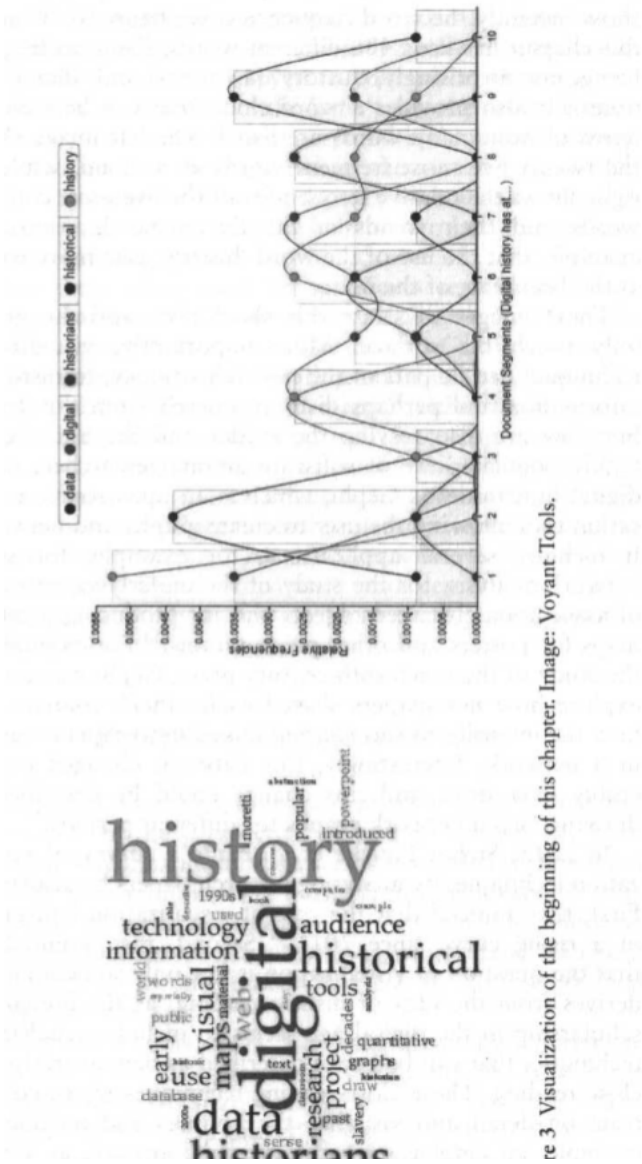
对Moretti而言，可视化是一种综合计算方法所得观察结果的手段。2005年出版的《Graphs, Maps, Trees: Abstract Models for a Literary History》中，他明确提出了这一点。Moretti认为，“有意识的缩减与抽象”是必要的；受其他研究领域启发，他采用了“来自量化史学的图表、来自地理学的地图，以及来自进化理论的树状图”。[8] 当然，图表一直是采用量化方法的经济史与社会史学者的重要工具。对统计专业知识的依赖在20世纪60至70年代尤其受到历史学家的重视。从这个意义上说，许多量化史学者或许会认为，较早的量化传统与新的数字史之间存在某种亲缘关系。不过，对Moretti来说，两者仍有不同，尤其体现在他如何把统计方法服务于“抽象”这一目标。

Statistical tools are necessary for a digital historian already because of the need to assess the data at hand. The question arises: how can the proportions of the data sets that are going to be used for research be described and how can they be illustrated clearly for the audience? In the study of big data, the researcher probably must first describe the data set in numbers. For example, in the case of text corpora, this involves how many characters, words, lines, pages, issues or books there are; the word frequencies in the data set; and perhaps the most common word collocations, that is, sequences of words that tend to co-occur in the material.[9] These proportions already help the reader to grasp quickly the size and quality of the material. This does not require visualization, though, and it can well be presented in textual and numerical form. There are easy-to-use tools for counting these properties, such as AntConc, which is a freeware toolkit for text corpus analysis.[10] For visualization, there are also open-source programs that can be easily tested. One of the most popular is Voyant Tools, a web-based application for text analysis and visualization.[11] To illustrate, the text of this chapter so far has been uploaded to Voyant Tools to show instantly the word frequencies (see figure 3). Thus far, this chapter has used 486 different words, the most frequent being, not surprisingly, ‘history’ (18 times) and ‘digital’ (17 times). It also provides a word cloud that can be scaled in terms of how many words are listed. The left image shows the twenty-five most frequent words as a cloud, while the right shows the relative frequencies of the five most common words and their trends in the document. It seems, for example, that the use of the word ‘history’ gets more weight at the beginning of the text.

由于需要评估手头的数据，统计工具对数字史研究者而言本身就十分必要。随之而来的问题是：研究将使用的数据集应当如何描述其规模和比例，又该如何清楚地向读者展示？在大数据研究中，研究者往往首先需要用数字描述数据集。例如，对于文本语料库，这意味着要说明其中有多少字符、词语、行、页、期刊期次或书籍；数据集中的词频如何；以及最常见的词语搭配是什么，也就是哪些词序列经常在材料中共同出现。[9] 这些比例信息已经可以帮助读者迅速把握材料的规模与性质。不过，这些信息未必一定要可视化，用文字和数字同样可以很好地呈现。现在已经有许多容易使用的工具可以统计这些属性，例如用于文本语料分析的免费工具包AntConc。[10]

在可视化方面，也有不少开源程序可供直接尝试。其中最流行的工具之一是Voyant Tools，这是一种基于网络的文本分析与可视化应用。[11]

作为示例，作者把本章到目前为止的文本上传到Voyant Tools，以便立即显示词频（见图3）。截至这里，本章共使用了486个不同的单词，其中出现最频繁的并不意外，是“history”（18次）和“digital”（17次）。Voyant Tools还可以生成词云，并可调整其中显示的词语数量。图中左侧显示最常见的25个词构成的词云，右侧则显示5个最高频词的相对频率以及它们在文档中的变化趋势。例如可以看到，“history”一词在文章开头部分所占的权重更大。



re 3 Visualization of the beginning of this chapter. Image: Voyant Tools.

Figure 3 Visualization of the beginning of this chapter. Image: Voyant Tools.

图3 本章开头部分的可视化。图片：Voyant Tools。

These images illustrate this short text, and the graphs only touch the surface. More importantly, visualization techniques can be part of the research strategy, to distil new information and perhaps distance oneself from the details, but they are also serving the reader and the audience. A highly popular piece of software among researchers in the digital humanities is Gephi, which is an open-source exploration tool allowing the user to create graphs and networks. It includes several applications, for example, for social network analysis, for the study of the underlying structures of associations between objects and for producing printable maps for posters and other presentations.[12] For example, in the study of the nineteenth-century press, Gephi was used to explore how newspapers shared each other's contents and how the intensity of this sharing linked newspapers together in a network. Interestingly, this network changed considerably over time, and this change could be described by drawing Gephi network graphs for different periods.[13]

这些图像只是用这个短文本做出的简单示例，而图表所呈现的也只是表层。更重要的是，可视化技术本身可以成为研究策略的一部分，用来提炼新的信息，并让研究者在一定程度上从细节中抽离出来；与此同时，它也服务于读者和观众。数字人文研究者中很受欢迎的一款软件是Gephi，它是一种开源探索工具，可以帮助使用者创建图表和网络。它可用于多种任务，例如社会网络分析、研究不同对象之间关联的深层结构，以及制作可用于海报和其他展示的网络图。[12] 例如，在19世纪报

刊研究中，研究者使用Gephi考察不同报纸如何相互转载内容，以及这种分享的强度如何把不同报纸连接成网络。值得注意的是，这个网络会随着时间发生显著变化，而这种变化可以通过为不同时期分别绘制Gephi网络图来呈现。[13]

In 2015, Stefan Jänicke et al. made a survey of visualization techniques by analysing research papers by academics. First, they noticed that the role of visualization had grown in a rising curve since 2011.[14] Second, they pointed out that the question of visualization is not only something that derives from the idea of distant reading, as the humanities scholarship in the digital age seems to include visualization techniques that can be better described as demonstrations of close reading. These close-reading techniques try to concentrate on detail and visualize the findings and include, for example, an emphasis on colour and font size, as well as on the use of glyphs and the aim to indicate connections between details. The distant-reading techniques in visualizations in turn emphasize structures and draw heat maps, tag clouds, timelines, maps and graphs. While the former techniques are often employed in connection to single text or parallel text analysis, in the latter case, the research often deals with large corpora, where spatial and temporal dimensions also have to be captured in the visualization.[15]

2015年，Stefan J nicke等人通过分析学术研究论文，对可视化技术进行了综述。首先，他们发现自2011年以来，可视化的重要性呈明显上升趋势。[14] 其次，他们指出，可视化并不只是源自“远读”理念；数字时代的人文学术同样包含一些更适合被理解为“细读展示”的可视化技术。这些细读技术强调细节，并把研究发现视觉化，例如突出颜色和字体大小、使用图形符号（glyph），以及显示不同细节之间的联系。相比之下，远读式可视化更强调结构，通常会使用热力图、标签云、时间线、地图和图表。前一类技术经常与单一文本或平行文本分析结合使用；后一类研究则往往面对大型语料库，因此还必须在可视化中表现空间和时间维度。[15]

The article further noted that the digital humanities community created visualizations ‘with the help of standard technologies such as HTML, JavaScript, SVG, or GIS’, but existent toolkits were often used ‘to transform textual data into visual metaphors’.[16] These toolkits included, for example, D3, Prefuse and ManyEyes.[17] For graphs, Gephi was one of the most popular, and D3, Neatline and GeoTemCo were the most popular for geographical maps.[18] Other toolkits were InfoVis, FeatureLens and TextArc.[19] Since this survey was carried out in 2015, the array of available libraries and software has become much larger. Many informative, curated websites have introduced useful tools for research, including many for visualization.[20] All this shows the need for visualization in the digital humanities and certainly in digital history too, not only as part of the research process but also in how to present the results to the audience.

该文还指出，数字人文社群会“借助HTML、JavaScript、SVG或GIS等标准技术”创建可视化，但研究者也经常使用现成工具包，“把文本数据转化为视觉隐喻”。[16]
这些工具包包括D3、Prefuse和ManyEyes等。[17] 对图表而言，Gephi是最常用的软件之一；对于地理地图，D3、Neatline和GeoTemCo则最为流行。[18]
其他工具还包括InfoVis、FeatureLens和TextArc。[19] 自2015年这项综述发表以来，可用的软件库和程序种类已经大幅增加。许多经过整理的信息网站也提供了研究工具，其中相当一部分用于可视化。[20] 这一切都说明，可视化对数字人文——当然也包括数字史——十分重要：它不仅是研究过程的一部分，也关系到研究者如何把结果呈现给受众。

Presenting change over time

呈现时间中的变化

The previous discussion on visualization is quite limited and technical from the perspective of how to present the past. While the visualization of data sets is valuable for the digital historian in communicating his/her research results, it is as important to explore solutions for crystallizing temporal change, particularly when presenting overarching historical interpretations for the audience. How to capture gradual change over time?

从“如何呈现过去”的角度来看，前面的可视化讨论仍然比较有限，而且偏技术性。数据集可视化固然有助于数字史研究者传达研究结果，但如何清晰呈现时间中的变化同样重要，尤其是在向公众展示总体性的历史解释时。问题是：我们应当怎样捕捉随着时间逐渐发生的变化？

Many of the tools developed in the digital humanities are tailored for statistical presentations or for describing the state of affairs at a particular point in history. Therefore, many historical surveys have offered graphs and charts that show how the research theme has developed on a temporal axis. It is also possible to release the findings as a database, where the user can adjust the time frame and, by these means, study how different points in time differ from each other.

数字人文中开发的许多工具，主要是为统计展示或描述某一历史时点的状态而设计的。因此，很多历史研究会使用图表来表现研究主题如何沿时间轴发展。研究成果也可以作为数据库发布，让使用者自行调整时间范围，并借此比较不同时点之间的差异。

Video presentations have also been employed to visualize change. *Viral Texts* has released several online videos that show how a particular text has travelled in time through republications by nineteenth-century US newspapers. The project includes, for example, a video on the spread of Charles MacKay’s poem ‘The Inquiry’ in pre-Civil War papers[21] and the viral expansion of Charles M. Dickinson’s popular poem ‘The Children’ that was, as soon as it appeared, attributed to Charles Dickens.[22] The project *Viral Culture in Early Nineteenth-Century Europe* has, in turn, released a video on Franz Liszt’s concert tours in Europe between 1839 and 1847,[23] while the English YouTuber Ollie Bye has published several videos on history - even the history of the world in 19 minutes.[24] In all these cases, time and space are inextricably entangled, and the temporal proceeding becomes visualized as changing overlays on a geographical map. A problem in these larger video visualizations might be that they tend to present temporality in a linear way, as a straightforward development from one point in time to another.

视频演示也被用来把变化过程视觉化。Viral Texts项目发布了多段在线视频，用19世纪美国报纸对某一文本的反复转载，展示它如何随着时间传播。例如，项目制作了Charles MacKay诗作《The Inquiry》在美国内战前报纸中传播的视频，[21] 也展示了Charles M. Dickinson的流行诗《The Children》如何迅速扩散——这首诗一出现就曾被误认为是Charles Dickens的作品。[22] 另一个项目《Viral Culture in Early Nineteenth-Century Europe》则制作了关于Franz Liszt在1839至1847年间欧洲巡演的视频；[23] 英国YouTuber Ollie Bye也发布了多部历史视频，甚至包括一部“19分钟世界史”。[24] 在这些例子中，时间与空间密不可分，时间的推进被表现为地理地图上不断变化的图层。不过，这类大型视频可视化也存在一个问题：它们往往把时间呈现为线性的，好像历史只是从一个时点直接发展到下一个时点。

Videos simulate the course of time, often by condensing timescales. It is, of course, also possible to try to convert temporality into spatial dimensions, which is the case in different timeline applications. To this end, the Knight Lab at Northwestern University has developed TimelineJS, an open-source tool that enables the user to build interactive timelines. Through a website interface, the user can add his/her own data as a Google spreadsheet or JSON file, and TimelineJS produces a webpage that shows a timeline with additional texts, images, videos and whatever the user wants to add on the line. The software is available in 60 languages, and it has been used by major media companies around the world, from the French newspaper *Le Monde* to the television channel CNN.[25]

视频通常通过压缩时间尺度来模拟时间流逝。当然，人们也可以尝试把时间性转换为空间维度，这正是各种时间线应用所做的事情。为此，美国西北大学Knight Lab开发了TimelineJS，这是一款可用于制作交互式时间线的开源工具。使用者可以通过网页界面，把自己的数据以Google表格或JSON文件形式导入，TimelineJS随后会生成网页时间线，并可在时间线上添加文字、图片、视频以及使用者希望加入的其他材料。该软件支持60种语言，也被世界各地大型媒体采用，从法国《Le Monde》到CNN电视台都有使用。[25]

Timeline applications are easy to use, but they also deserve critical attention. In historical research today, the question of different, overlapping rhythms in time have aroused increasing interest.[26] If the past is presented as a timeline, its focus has to be clearly explicated, since, in the past, there never was only one timeline. Merely, history consists of multiple temporalities that exist simultaneously.

时间线应用虽然容易使用，但也需要受到批判性审视。当代历史研究越来越重视不同时间节奏彼此交叠的问题。[26] 如果用时间线来呈现过去，就必须清楚说明这条时间线关注的是什么，因为真实的过去从来不只有一条单一时间线。历史实际上由多种同时存在的时间性构成。

Augmented and mixed realities

增强现实与混合现实

The aforementioned examples could be enriched by the kinds of public history projects that draw on recent developments in digital technologies and use visualization to invoke interest among audiences, particularly by emphasizing the experientiality of history, the sense of living in the past. One of the most vibrant areas of research and development emerged in the 1990s under the umbrella of virtual reality (VR), the effort to create immersive artificial digital environments. If these environments were still rather rough in the 1990s, today VR is a sophisticated field of cultural production, used in gaming but also in many practical matters, such as pilot training. Coined by Tom Caudell in 1990, the notion of augmented reality (AR) was built on VR. AR means that the perception of the observer is fed with virtual objects that are overlaid on top of the real-world environment. Mixed reality (MR) goes a bit further not only by overlaying digital objects but also by anchoring them to the real world; thus, virtual and physical objects can coexist in real time. In this sense, MR aims at presenting a hybrid between the physical and virtual worlds.[27]

前面提到的这些做法，还可以通过近年数字技术发展的成果进一步丰富。公共史项目能够利用可视化吸引观众，尤其是通过强调历史的体验性，让人产生“生活在过去”的感觉。20世纪90年代，虚拟现实（VR）成为研究与开发中最活跃的领域之一，其目标是创造沉浸式的人工数字环境。虽然90年代的虚拟环境还比较粗糙，但今天VR已经成为成熟的文化生产领域，不仅用于游戏，也应用于飞行员训练等许多实际场景。Tom Caudell在1990年提出“增强现实”（AR）概念，它建立在VR基础之上。AR是指在观察者看到的真实世界环境上叠加虚拟对象。混合现实（MR）则更进一步：它不仅叠加数字对象，还把这些对象锚定在现实世界中，使虚拟对象与物理对象能够在实时环境中共存。从这个意义上说，MR试图呈现物理世界与虚拟世界之间的混合状态。[27]

Therefore, the goal of MR is to fuse elements of the real and digital worlds, offering a sense of presence. In digital history projects, using MR can be a method to express temporal distance and affinity at the same time by enabling the mixture of different temporal layers and historical periods; thus, it can be employed for many purposes - pedagogical, touristic or edutainment. This technology, as Jennifer Challenor and Minhua Ma write, ‘has rapidly become more viable for commercial and research projects in the last decade due to the prevalence of head-mounted devices (HMDs) and smart devices such as phones, tablets and handheld games consoles that

are now intrinsically woven into daily life'.[28] The most popular AR/MR apps are games, and the success of Pokémon GO in 2016 made it visible and known to millions of people worldwide. MR applications have also been successful in museums and touristic activities, as well as in history classes, and there is an ever-expanding academic literature on the challenges and possibilities of this technology for educationalists.

因此，MR的目标是融合现实世界与数字世界的元素，并创造在场感。在数字史项目中，MR可以把不同的时间层次和历史时期混合起来，从而同时表达时间上的距离感与亲近感；因此，它可以服务于教学、旅游或寓教于乐等多种目的。正如Jennifer Challenor和Minhua Ma所写，过去十年间，随着头戴式设备（HMD）以及手机、平板电脑和掌上游戏机等智能设备普及并深深嵌入日常生活，这项技术“迅速变得更适合商业和研究项目”。[28]
最受欢迎的AR/MR应用主要是游戏，2016年Pokémon GO的成功让这种技术被全球数百万人认识。MR也成功应用于博物馆、旅游活动和历史课堂，关于这项技术对教育工作者所带来的挑战与可能性的学术研究也在不断增长。

AR applications are often connected with the idea of immersive learning: they aim at adding a playful element to history teaching and giving an interactive impetus for learning.[29] Many of these applications are commercial products for the classroom, including apps on historical material objects, such as an Egyptian sarcophagus or a Roman ship, or environments, such as London in 1666.[30] Museums have also launched large-scale AR installations. The National Museum of Singapore has opened an exhibition titled *Story of the Forest* in its Glass Rotunda, created by the Japanese digital art collective teamLab. The immersive installation is based on sixty drawings from the William Farquhar Collection of Natural History Drawings, which have been transformed into 3D animations. The exhibition delves into the flora and fauna of Singapore, as well as into colonial past and present-day concerns. The visitor can download a free app before entering and learn more via the AR elements added to the exhibition.[31]

AR应用经常与“沉浸式学习”理念联系在一起：它们试图给历史教学加入游戏化元素，并通过互动增强学习动力。[29] 其中许多是面向课堂的商业产品，包括围绕历史物质对象开发的应用，例如埃及石棺或罗马船只；也包括重建特定环境的应用，例如1666年的伦敦。[30]
博物馆也开始推出大型AR装置。新加坡国家博物馆就在Glass Rotunda中开放了名为《Story of the Forest》的展览，由日本数字艺术团队teamLab创作。这个沉浸式装置以William Farquhar Collection of Natural History Drawings中的60幅图画为基础，把它们转化为3D动画。展览既探索新加坡的动植物，也涉及殖民历史和当代议题。参观者可以在进入展览前下载免费应用，并通过展览中加入的AR元素获得更多信息。[31]

AR and MR have many applications for digital historians. They provoke theoretical questions on the role of historical knowledge in the construction of immersive worlds where past and present do not only meet but can be put into a dialogue with each other.[32] AR and MR can also be created in a way that they simultaneously encourage social engagement and entail research questions to be answered. MR has been successfully employed, for example, in offering historical reconstructions of lost or destroyed architectural surroundings via digital technology.[33] In 2017, the MIRACLE project produced several demonstrator applications that were used in historical sites and museums. The project also produced a guidebook for mixed reality applications for cultural and learning experiences, including an analysis of both the benefits and risks of MR.[34]

AR和MR对数字史研究者有许多用途。它们也提出了理论问题：在构建沉浸式世界时，历史知识究竟发挥什么作用？在这些世界中，过去与现在不仅相遇，还可以彼此形成对话。[32] AR和MR还可以被设计成一方面鼓励社会参与，另一方面把尚待回答的研究问题带入其中。例如，MR已经成功用于通过数字技术重建已经消失或被毁坏的建筑环境。[33] 2017年，MIRACLE项目制作了多种示范应用，并把它们用于历史遗址和博物馆。该项目还编写了一本关于混合现实在文化与学习体验中应用的指南，其中同时分析了MR的优势与风险。[34]

Travelling in time

穿越时间

In recent years, large overarching projects have been launched to strengthen digital history in research, teaching and social outreach. One of the most ambitious ventures is the European-wide project *Time Machine* that aims to implement the mass digitization of historical collections with smart scanning technologies, to bring together resources that are now scattered around the continent, to apply AR and MR applications as well as, and, above all, to highlight the potential of the 2000-year history of European culture.[35] The project was inspired by the earlier Venice Time Machine, launched by the École Polytechnique Fédérale de Lausanne and the Ca' Foscari University of Venice in 2012, with the idea to produce a multidimensional digital model of Venice that could also serve as an access point to the open digital archive of the city.[36] The time machine is, of course, a powerful metaphor. It does not involve time travel, in the spirit of H.G. Wells, but supports the idea that digital technologies can provide tools, for all users, to explore history and engage in a fruitful and thought-provoking dialogue between the present and the past.

近年来，一些大型综合项目相继启动，以加强数字史在研究、教学和社会传播中的作用。其中最具雄心的计划之一，是覆盖欧洲的《Time Machine》项目。它希望借助智能扫描技术大规模数字化历史收藏，把目前散落在欧洲各地的资源汇集起来，同时应用AR与MR，并最重要地展示欧洲两千年文化历史所具有的潜力。[35] 这一项目受到更早的Venice Time Machine启发。后者于2012年由洛桑联邦理工学院与威尼斯Ca' Foscari大学启动，目标是建立一个多维的威尼斯数字模型，同时把它作为进入城市开放数字档案的入口。[36]

"时间机器"当然是一个强有力的隐喻。它并不是H.G. Wells式的真正时间旅行，而是强调数字技术能够为所有使用者提供探索历史的工具，并促使当下与过去之间形成富有成果、能够激发思考的对话。

Public history has been an essential ingredient in digital history since the 1990s, and it continues to be so with new technologies and new research settings. The presentation of history also involves ethical commitments of which we are increasingly aware. History is an active mode of knowledge and belongs to everyone, and therefore, historians are expected to converse with both other researchers and the general public. Today, historians publish their research through open access publications but also through digital technologies that allow an array of narrative possibilities, from blog posts to databases, from social media updates to curated services. For this continuing work, digital history provides tools and insights to engage with our time.

自20世纪90年代以来，公共史一直是数字史不可或缺的组成部分；随着新技术和新研究环境出现，这一点仍然没有改变。历史的呈现还涉及我们越来越重视的伦理责任。历史是一种积极的知识形态，也属于所有人，因此历史学家不仅应与同行交流，也应与普通公众对话。今天，历史学家可以通过开放获取出版物发表研究，也可以借助数字技术采用多种叙事形式——从博客文章到数据库，从社交媒体更新到经过策划的数字服务。对于这项持续进行的工作，数字史为我们提供了理解并参与自身时代的工具与洞见。